By submitting this assignment, I agree to the following:
"Aggies do not lie, cheat, or steal, or tolerate those who do."
"I have not given or received any unauthorized aid on this assignment."
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Section: 571
Assignment: Project Planning - Submit the Project Hierarchy
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Hierarchy

- Menu
    - Print out menu with list of games
    - Get input from user
        - if input is 1, then rock paper scissors
        - if input is 2, then tic-tac-toe
        - if input is 3, then hangman
- Rock paper scissors
    - Game loop will be a while true loop
        - Get choice from user
            - Get input from user
            - If input is “exit”, then exit out of the game and return to menu
        - Get choice from computer
            - Random choice between rock, paper, and scissors
        - Determine outcome of the game
            - Rock beats scissors, paper beats rock, and scissors beats paper.
            - If the choices are the same then it is a draw.
        - Display winner, or draw, and a message to try again or exit
- Tic-tac-toe
    - Make dictionary to form display
    - Function to display tic-tac-toe grid
        - Use dictionary to convert “X” or “O” to ascii
    - Game loop will be a while true loop
        - Initialize 3x3 list with empty strings
        - Initialize a boolean variable that represents whether it is the first player turn.
        - Display tic-tac-toe grid
        - Turn loop will be a while true loop
            - Get position from user
                - Get input from user
                - If input is “exit”, return to main menu
                - If input is “restart”, restart the game
                - Check if input is valid
                    - Position entered must exist

                        - Position entered must be available (is empty string)
                        - Input must have x and y and should be in (x,y) coordinate system format
                        - Must be number
                    - If not valid, input display message to try again
                - Update 3x3 list
                    - If player 1, add “X” in that spot
                    - If player 2, add “O” in that spot
                - Display tic-tac-toe grid
                - Check if someone wins
                    - Check for horizontal win
                    - Check for vertical win
                    - Check for diagonal win
                    - If there is a winner
                        - Print that there is a winner
                        - Break out of turn loop
                    - If there is no winner, continue the turn loop
                - Check if matrix is full
                    - Print that it is a tie
                    - Break out of turn loop
            - Call out result message to try again or exit
                - “Try again” to start over
                - “Exit” to return to menu
- Hangman
    - Game loop will be a while true loop
        - Get the hidden word from user
            - Store the hidden word or phrase in a variable
            - Convert the hidden word to uppercase
        - Store a string representing the current guesses, called current_word
            - String has same length as hidden word or phrase
            - Characters that are not spaces are represented with underscore
        - Guess loop will be a while true loop
            - User inputs letter until valid
                - Get input from user
                - Check if input is valid
                    - If input is “exit”, exit out of the game
                    - If input is “restart”, restart the game
                    - If input is not one letter, prompt the user to enter another guess
                    - If input is one letter
                        - Check if letter has already been guessed
                            - If already guessed, prompt the user to enter another guess

                - If letter has not been guessed, exit out of the loop
        - Check the guess
            - Convert the letter to uppercase
            - Check if letter in word
                - If the guessed letter is in the word
                    - Update current_word so that all places where the letter is is set to that letter
                    - Add letter to list of correct letters
                - If not in word, add letter to list of incorrect letters
        - Display hangman
            - Each number of incorrect guesses corresponds to a body part
                - 1 for head, 2 for neck, 3 for torso, 4 for left arm, 5 for right arm, 6 for left leg, 7 for right leg
            - Print hangman based on number of incorrect guesses
        - Print current_word
        - If the user has 7 incorrect guesses
            - Print to the user that they lost
            - Break out of guess loop
        - Check if word is complete
            - If all letters of word have been guessed
                - Print that the game is complete
                - Break out of guess loop
            - If not, continue the game loop
- Get input from user if they want to play again
    - If input is “yes”, restart the game
    - If input is “no”, exit out of the game and return to menu